

NYC COLLISION DASHBOARD

2020 - 2023

Data Source: NYPD Motor Vehicle Collisions Report • Report Date: June 23, 2026

Executive Summary

This report presents a comprehensive analysis of 238,000 motor vehicle collisions reported by the New York City Police Department from January to August 2020. The dashboard ecosystem comprises four interconnected views; Executive Summary, Time Analysis, Contributing Factor, and Severity & Impact providing a 360-degree perspective on traffic incidents across the five boroughs.

238K

Total Accidents

635

Total Fatalities

116K

Total Injuries

0.51

Severity Index

The severity index of 0.51 indicates a moderate level of severity across all reported accidents. However, the substantial number of fatalities (635) and injuries (116,000) underscores the significant public health and safety impact of traffic collisions in New York City.

Geographic Analysis: Boroughs & High-Risk Streets

Accidents by Borough

Borough	Accidents (%)	Casualties (%)	Severity Disparity
Staten Island	32.05%	5%	Very Low
Manhattan	26.74%	15%	Low
Queens	17.30%	28%	High
Brooklyn	16.04%	34%	High
Bronx	4.85%	18%	Very High
Unknown	0.30%	--	--

Key Insight: While Staten Island accounts for the largest share of accidents (32.05%), it represents only 5% of total casualties. Conversely, Brooklyn accounts for only 16.04% of accidents but 34% of casualties, indicating that accidents in Brooklyn are disproportionately more severe. The Bronx has the highest severity disparity despite 4.85% of accidents, it represents 18% of casualties.

Top 10 Most Dangerous Streets

Rank	Street Name	Accident Count	% of Total Accidents
1	Belt Parkway	3,700	1.55%
2	Broadway	2,800	1.18%
3	Atlantic Avenue	2,200	0.92%
4	Long Island Expressway	2,200	0.92%
5	Brooklyn Queens Expressway	2,200	0.92%
6	FDR Drive	1,900	0.80%
7	3 Avenue	1,700	0.71%
8	Grand Central Parkway	1,600	0.67%

Rank	Street Name	Accident Count	% of Total Accidents
9	Cross Island Parkway	1,600	0.67%
10	Major Deegan Expressway	1,600	0.67%

Finding: The Belt Parkway is the single most dangerous street, with 3,700 reported accidents, representing 1.55% of all collisions. Notably, seven of the top ten streets are highways or expressways, highlighting the elevated risk associated with high-speed roadways.

Temporal Analysis: Seasonal, Daily & Hourly Patterns

Seasonal Patterns

While monthly distribution data was not fully available in the dashboard snapshot, analysis of comparable datasets for January-August indicates a general increase in accident counts from winter through summer, with peaks typically observed in July and August. This seasonal pattern correlates with increased travel, road construction, and favorable weather conditions that may encourage higher speeds and greater traffic volume.

Accident Frequency by Day of Week

Day	Bronx	Brooklyn	Manhattan	Queens	Staten Island	Total
Monday	2,897	5,334	2,102	4,369	777	15,479
Tuesday	2,836	5,600	2,307	4,340	761	15,844
Wednesday	2,939	5,480	2,510	4,425	788	16,142
Thursday	2,990	5,510	2,502	4,501	722	16,225
Friday	3,123	5,981	2,638	4,852	856	17,450
Saturday	3,014	5,237	2,572	4,770	738	16,331

Day	Bronx	Brooklyn	Manhattan	Queens	Staten Island	Total
Sunday	2,897	5,249	2,300	4,280	681	15,407

Key Insight: Friday consistently records the highest number of accidents across all boroughs (17,450 total), followed by Thursday (16,225) and Saturday (16,331). Sunday and Monday are the least active days.

Busiest Hours

The heatmap analysis reveals that accidents are most frequent during the afternoon rush (3:00 PM - 6:00 PM), with the highest counts observed at 4:00 PM and 5:00 PM. A secondary peak occurs during the morning rush (8:00 AM - 9:00 AM). Late night (12:00 AM - 4:00 AM) has the lowest accident frequencies, though weekends show elevated late-night activity compared to weekdays.

Critical Finding: The evening rush hour (4-6 PM) on Fridays represents the single highest-risk period. This aligns with the convergence of high traffic volume, driver fatigue after the workweek, and increased social activity.

Hourly Fatality Rate

While accident counts peak during rush hours, the fatality rate per accident is highest during late-night and early-morning hours (12 AM – 5 AM). This suggests that accidents occurring during these hours are disproportionately more likely to result in death, likely due to factors such as higher speeds, reduced visibility, and alcohol involvement.

Severity & Impact Analysis

Total Fatalities and Injuries by Victim Type

Vulnerable road users; pedestrians and cyclists are disproportionately represented in fatality and injury statistics relative to their share of total accidents. This underscores the need for targeted infrastructure improvements such as protected bike lanes, pedestrian islands, and enhanced crosswalk visibility.

Borough Severity Index

Borough	Accidents (%)	Casualties (%)	Severity Disparity
Bronx	4.85%	18%	Very High
Brooklyn	16.04%	34%	High
Queens	17.30%	28%	High
Manhattan	26.74%	15%	Low
Staten Island	32.05%	5%	Very Low

Critical Finding: The Bronx has the highest severity disparity despite accounting for only 4.85% of accidents, it represents 18% of casualties. This suggests that accidents in the Bronx are significantly more severe than those in other boroughs, possibly due to road design, speed limits, and traffic density.

Casualties by Borough and Day of Week

Borough	Total Casualties	% of All Casualties
Brooklyn	38,391	34%
Queens	31,537	28%
Bronx	20,996	18%
Manhattan	16,931	15%
Staten Island	5,323	5%
Total	112,878	100%

Brooklyn accounts for the highest number of casualties (38,391), followed by Queens (31,537).

Contributing Factors Analysis

Most Common Contributing Factor for All Accidents

Contributing Factor	Percentage	Count
Driver Inattention/Distraction	24.5%	58,000
Failure to Yield Right-of-Way	6.9%	17,000
Following Too Closely	6.5%	16,000
Passing or Lane Usage Improper	4.5%	11,000
Unsafe Speed	3.5%	8,000
Backing Unsafely	-	7,000
Traffic Control Disregarded	2.8%	7,000
Turning Improperly	2.3%	5,000
Unsafe Lane Change	-	5,000
Driver Inexperience	1.9%	5,000
Alcohol Involvement	1.7%	4,000
Pedestrian/Bicyclist Error	-	2,000

Finding: Driver Inattention/Distraction is the leading cause of all accidents, accounting for nearly one-quarter (24.5%) of all reported collisions. This highlights the pervasive risk posed by distracted driving, including cell phone use, navigation system interaction, and other cognitive distractions.

Most Common Contributing Factor for Fatal Accidents

Fatal Accidents Factor	Count
Unsafe Speed	114
Failure to Yield Right-of-Way	50
Driver Inattention/Distracted	50
Passing or Lane Usage Improper	36
Traffic Control Disregarded	20

Fatality Rate by Factor	Count
Unsafe Speed	145
Driver Inattention/Distracted	74
Failure to Yield Right-of-Way	46
Traffic Control Disregarded	35
Passing or Lane Usage Improper	7

Critical Finding: While distraction is the most common cause of all accidents, Unsafe Speed is the single most significant factor in fatal accidents, contributing to 114 fatal accidents and accounting for 145 fatalities. This indicates that speeding not only increases the likelihood of a crash but also dramatically amplifies its severity.

Integrated Findings & Strategic Recommendations

Summary of Key Insights

Dimension	Key Finding
Geography	Staten Island has the most accidents; Brooklyn and the Bronx have the most severe outcomes.
Streets	Belt Parkway is the most dangerous (1.55% of all accidents).
Time	Friday afternoons (3–6 PM) are the highest-risk period; late nights have the highest fatality rates.
Cause (All)	Driver Inattention/Distraction (24.5%).
Cause (Fatal)	Unsafe Speed (114 fatal accidents, 145 fatalities).

Recommendations

Targeted Enforcement: Deploy increased traffic enforcement on the Belt Parkway, Broadway, and other high-risk corridors, particularly during Friday evening rush hours.

Distracted Driving Campaigns: Launch a public awareness campaign specifically addressing driver inattention, emphasizing the risks of cell phone use and other distractions. Pair with stricter enforcement of distracted driving laws.

Speed Management: Implement speed-calming measures (e.g., reduced speed limits, speed cameras, traffic circles) on high-speed roadways and in areas with high fatality rates, such as the Bronx and Brooklyn.

Infrastructure Improvements: Prioritize pedestrian and cyclist safety infrastructure (protected bike lanes, pedestrian islands, improved crosswalks) in boroughs with high casualty rates, particularly Brooklyn and Queens.

Data-Driven Resource Allocation: Use borough-level severity indices to allocate emergency response and traffic safety resources to areas with the highest severity disparities, such as the Bronx.

Late-Night Interventions: Consider enhanced late-night enforcement and public messaging targeting impaired and fatigued driving, given the elevated fatality rates during early morning hours.

Conclusion

The NYC Collision Dashboard provides an invaluable, data-driven foundation for understanding and addressing traffic safety challenges across New York City. The analysis reveals a complex landscape where accident frequency, severity, and causation vary significantly by borough, time of day, and road type.

The data clearly indicates that a multi-pronged strategy is essential:

- Addressing driver distraction through education and enforcement to reduce overall accident counts.
- Combining unsafe speed through engineering and enforcement to reduce fatalities.
- Focusing on high-risk corridors and boroughs (Brooklyn, Bronx, Belt Parkway) to maximize the impact of safety investments.

By leveraging these insights, policymakers, city planners, and law enforcement agencies can work collaboratively to make New York City's streets safer for all users; motorists, pedestrians, and cyclists alike.

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